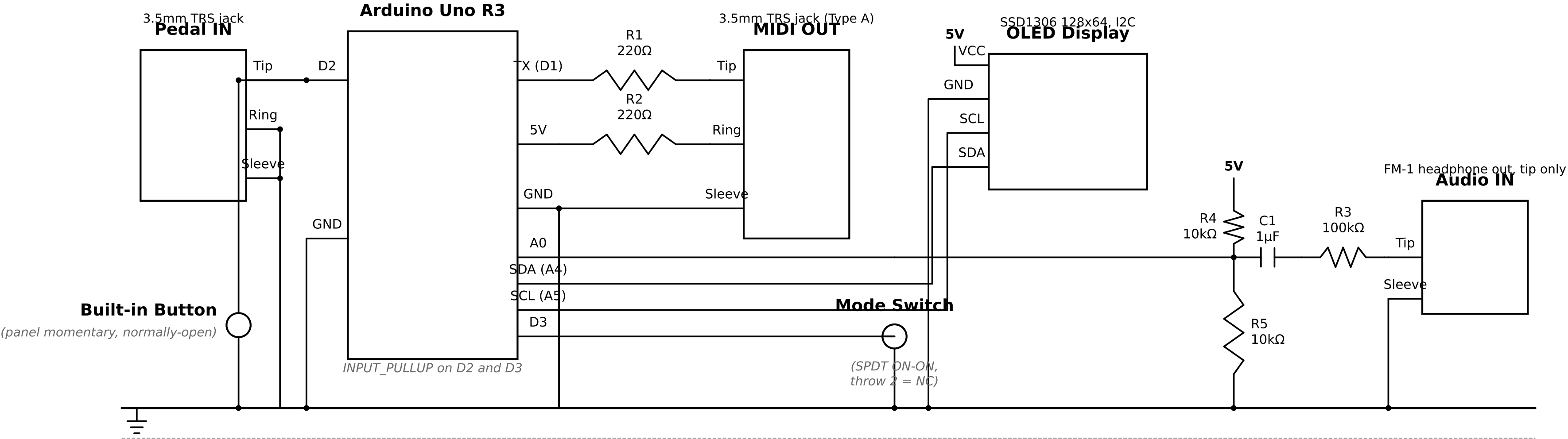


FM-1 Sustain Footswitch — Schematic

Arduino Uno R3 + TRS Pedal In + Built-in Button + TRS MIDI Out (Type A) + OLED Mode Switch + Pitch-Detect Audio In



Notes:

- Built-in Button is wired in parallel with the pedal jack (same D2/GND nodes) -- either one alone can trigger sustain, so an external pedal is optional
- Pedal ring + sleeve are tied together so a plain mono (TS) pedal plug still grounds correctly in the TRS jack
- D2 uses INPUT_PULLUP -- the button or pedal only needs to short D2 to GND when pressed (normally-open momentary)
- Type A TRS MIDI: tip = signal, ring = +5V current source, sleeve = ground
- R1 limits current through the FM-1's opto-isolated MIDI input; R2 sources the +5V loop current -- both 220Ω, matching the standard DIN MIDI-out circuit
- TX (D1) is shared with the USB-serial programmer -- disconnect the MIDI OUT jack (or at least the R1 lead) before uploading a new sketch
- If the FM-1 doesn't respond, it may expect Type B instead: swap Tip and Ring at the MIDI OUT jack (tip = GND, ring = signal, sleeve = GND)
- Mode Switch is optional -- D3 uses INPUT_PULLUP, so leaving it unwired (or on throw 2) defaults to Sustain mode, identical to the switch-less build
- Pitch mode reads the FM-1's own headphone output, not MIDI -- this board has no incoming MIDI connection at all
- R4/R5 bias A0 to ~2.5V so the AC headphone signal (which swings positive and negative around 0V) stays inside the ADC's 0-5V range
- C1 blocks the DC bias from reaching the FM-1; R3 limits current and knocks down the signal as cheap insurance against clipping at high FM-1 volume
- Pitch detection (autocorrelation, in-sketch) is tuned for roughly 50Hz-1000Hz (about G1-B5) -- a practical range for one ATmega328p doing this in real time, not the FM-1's full range
- Untested against a real FM-1 signal as of this diagram -- the sketch's noise-floor and confidence thresholds are starting points and will likely need tuning